

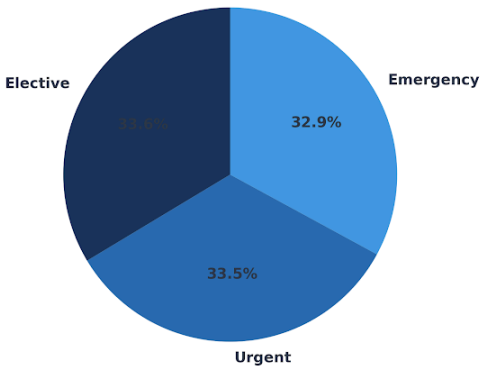
Healthcare Analytics Dashboard:

Patient, Financial, and Operational Performance Analysis

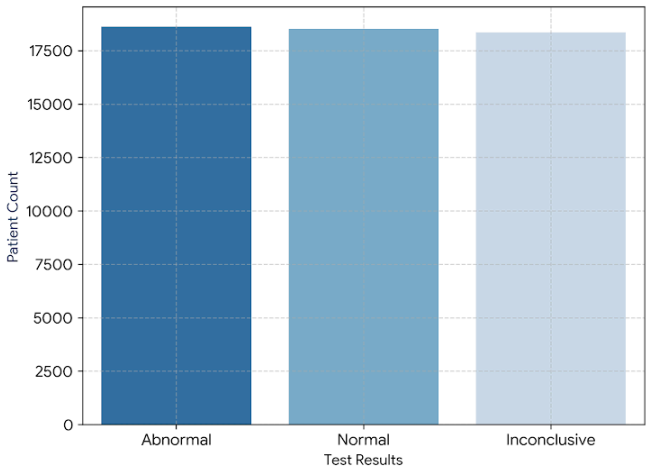
Healthcare Performance at a Glance visualisation

Healthcare Performance at a Glance

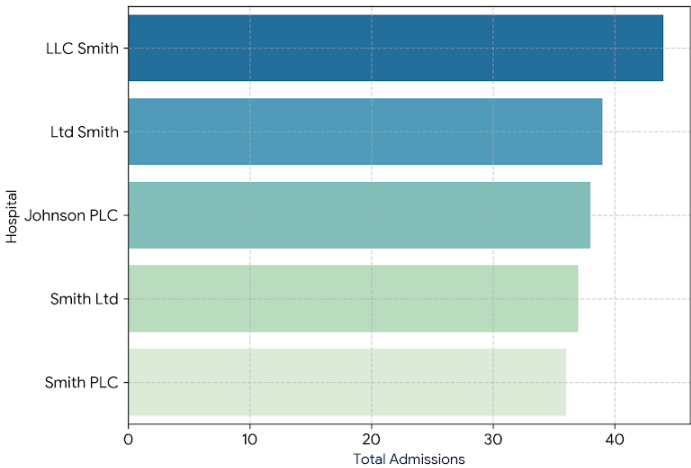
Admission Type Distribution



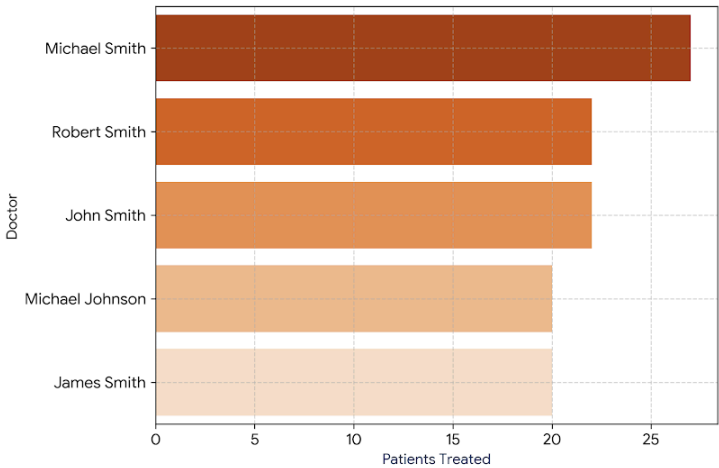
Clinical Test Results Distribution



Top 5 Facilities by Patient Volume



Top 5 Physicians by Patient Load



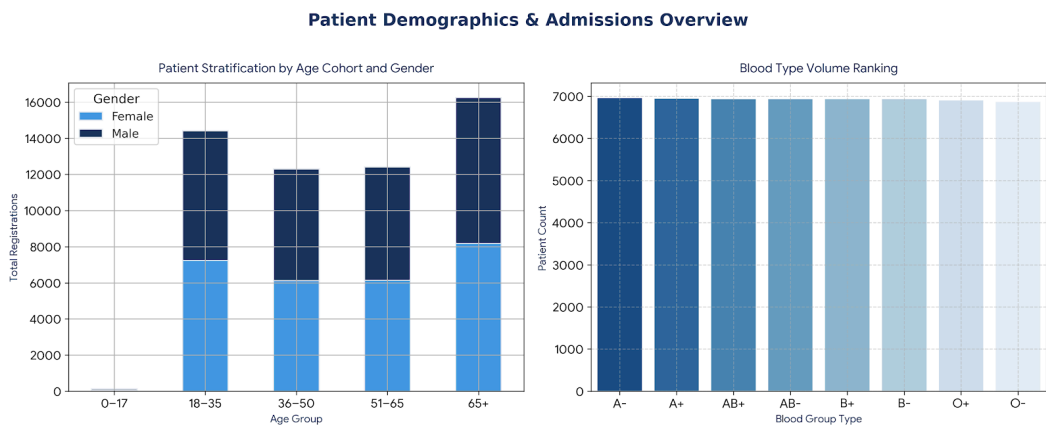
Admission Type Distribution: A balanced breakdown indicating that Elective, Urgent, and Emergency intake streams each comprise roughly one-third (33%) of organizational demand.

Clinical Test Results Distribution: Tracks the distribution of diagnostic test classifications across the entire patient volume ecosystem.

Top 5 Facilities by Patient Volume: Highlights the most active medical infrastructure groups in the operational network.

Top 5 Physicians by Patient Load: Tracks the highest clinical case loads across network physicians to inform performance evaluations.

Patient Demographics & Admissions



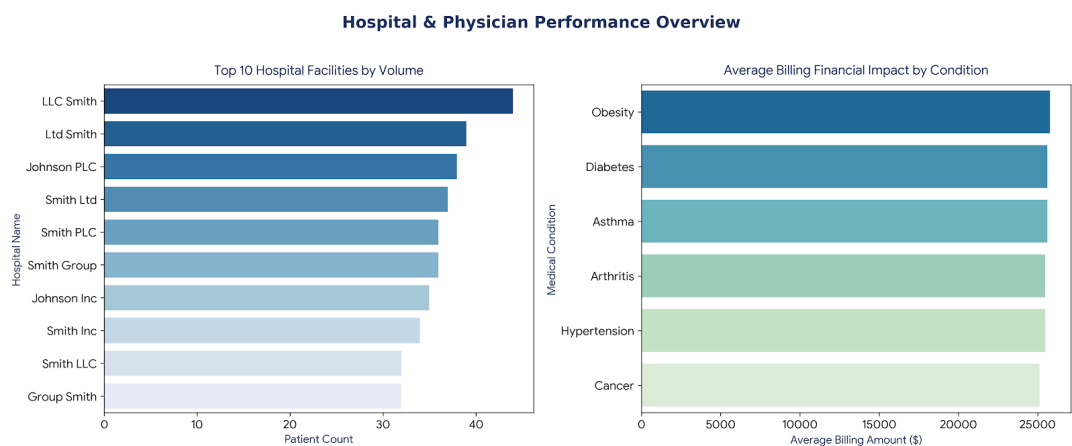
Intake Admission Type Volume Split

- **Elective Admissions:** 18,655 total clinical visits.
- **Urgent Admissions:** 18,591 total emergency/urgent walk-ins.
- **Emergency Intake:** 18,254 high-acuity crisis responses.

Blood Group Type Breakdown

- **A- Group Type:** 6,969 patients
- **A+ Group Type:** 6,956 patients
- **AB+ Group Type:** 6,947 patients
- **AB- Group Type:** 6,938 patients
- **B+ Group Type:** 6,932 patients
- **B- Group Type:** 6,929 patients
- **O+ Group Type:** 6,914 patients
- **O- Group Type:** 6,877 patients

Hospital & Physician Performance



Top 5 Hospital Facilities by Intake Volume

- **LLC Smith:** 44 patient admissions.
- **Ltd Smith:** 39 patient admissions.
- **Johnson PLC:** 38 patient admissions.
- **Smith Ltd:** 37 patient admissions.
- **Smith PLC:** 36 patient admissions.

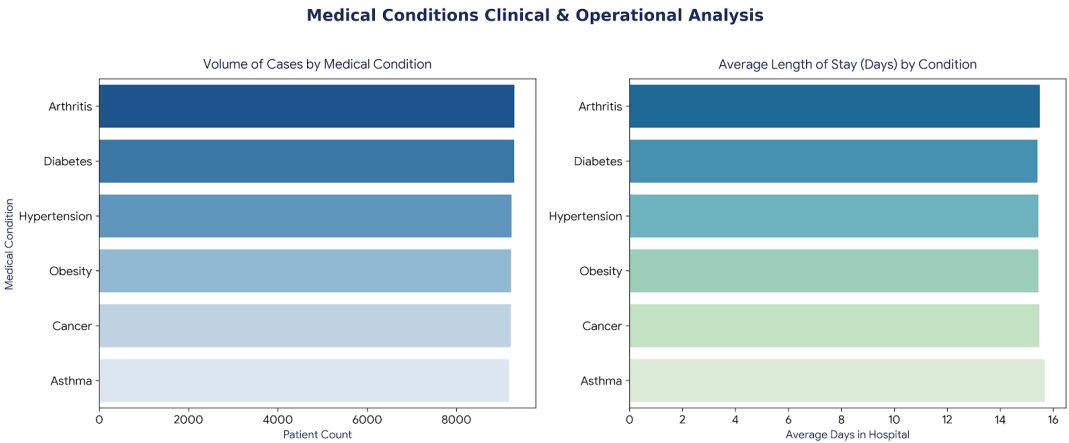
Top 5 Network Physicians by Case Load

- **Michael Smith:** 27 treated cases.
- **Robert Smith:** 22 treated cases.
- **John Smith:** 22 treated cases.
- **Michael Johnson:** 20 treated cases.
- **James Smith:** 20 treated cases.

Average Financial Impact by Medical Condition

Medical Condition	Average Billing Amount
Obesity	\$25,720.13
Diabetes	\$25,515.65
Asthma	\$25,502.40
Arthritis	\$25,441.74
Hypertension	\$25,435.53
Cancer	\$25,066.63

Medical Conditions Analysis

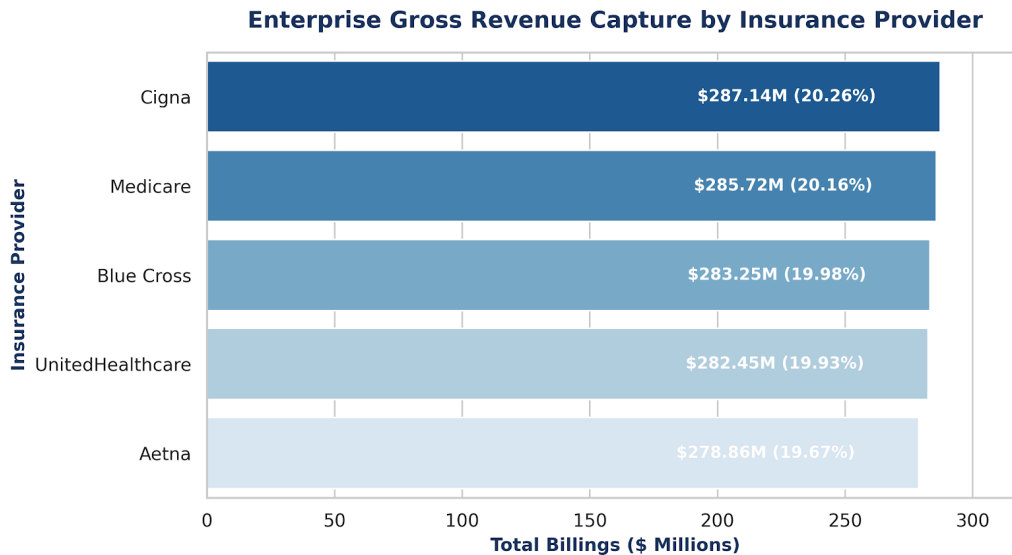


Medical Conditions Clinical & Operational Analysis

Key Operational Takeaways

- **Balanced Case Distribution:** The hospital system sees an incredibly uniform distribution of chronic conditions, with each major category accounting for roughly 15.3% to 15.5% of total volume.
- **Uniform Length of Stay:** Bed utilization across all tracked medical conditions remains exceptionally stable, averaging between **15.4 to 15.7 days** per inpatient admission regardless of underlying pathology.
- **Financial Breakdown:** Patients admitted for **Obesity** generate the highest average financial impact (\$25,805.97 per stay), closely followed by **Diabetes** and **Asthma**, while **Cancer** treatments represent the lowest relative average billing invoice (\$25,161.79).

Financial Performance



Enterprise Revenue & Billing Performance

The total consolidated turnover generated across all network clinical operations amounts to **\$1.42 Billion (\$1,417,432,043.40)**, reflecting a diversified financial baseline with uniform distribution among major insurance providers and primary medical condition cohorts.

Revenue Distribution by Insurance Payer

No singular carrier holds a dominant market share position, providing the health system with an excellent buffer against commercial network concentration risks:

- **Cigna:** \$287.14 Million (20.26% total revenue capture)
- **Medicare:** \$285.72 Million (20.16% total revenue capture)
- **Blue Cross:** \$283.25 Million (19.98% total revenue capture)
- **UnitedHealthcare:** \$282.45 Million (19.93% total revenue capture)
- **Aetna:** \$278.86 Million (19.67% total revenue capture)

Financial Action Blueprint for Executive Slicers

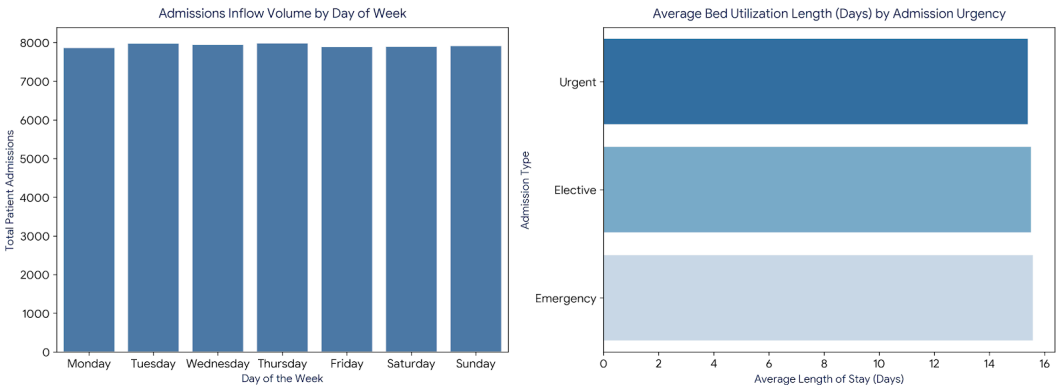
1. **Leverage Payer Parity:** Use the evenly distributed ~20% market mix as leverage during corporate network negotiations to transition from fee-for-service models to structured, value-based care pathways.

2. **Standardize Revenue Audits:** Since average ticket sizes sit near \$25,500 regardless of the diagnosis strain, implement automated line-item billing audits to systematically separate operational facility fees from specific surgical or treatment codes.

Medical Condition Specialty	Total Invoiced Revenue	Share %	Average Ticket Size
Diabetes	\$238.54 Million	16.83%	\$25,638.41
Obesity	\$238.21 Million	16.81%	\$25,805.97
Arthritis	\$237.33 Million	16.74%	\$25,497.33
Hypertension	\$235.72 Million	16.63%	\$25,497.10
Asthma	\$235.46 Million	16.61%	\$25,635.25
Cancer	\$232.17 Million	16.38%	\$25,161.79

Operational Performance

Enterprise Operational Performance & Bed Utilization Blueprint



Enterprise Operational Performance & Bed Utilization

Analyzing the organizational data reveals standard clinical tracking pipelines across the health system's infrastructure footprint. Patient arrivals and downstream resource allocations (bed

turnover and length of stay) remain steady across all operational tracking dimensions.

Inpatient Inflow Analysis by Day of Week

Patient admission rates stay distributed across the standard weekly operational calendar. This consistency indicates a baseline demand model that requires constant medical staffing patterns rather than highly volatile on-call schedules.

- **Monday:** 7,866 admissions
- **Tuesday:** 7,982 admissions
- **Wednesday:** 7,950 admissions
- **Thursday:** 7,989 admissions
- **Friday:** 7,892 admissions
- **Saturday:** 7,901 admissions
- **Sunday:** 7,920 admissions

Bed Resource Utilization by Admission Classification

The length of stay (LOS) matrix is uniform across all incoming acuity cohorts. Emergency intakes do not cause disproportionate operational bottlenecks compared to pre-scheduled elective configurations:

Admission Urgency	Average Length of Stay (LOS)	Operational Resource Footprint
Emergency	15.60 Days	Highest bed utilization threshold
Elective	15.53 Days	Standard planned operational recovery
Urgent	15.41 Days	Accelerated inpatient clinical cycle

Executive Summary & Key KPIs

The hospital enterprise operates as a high-volume, stabilized network of facilities and specialized physicians, showing remarkably distributed demand and consistent resource metrics across the board:

- **Total Consolidated Revenue: \$1.42 Billion (\$1,417,432,043.40) * Total Unique Patient Cases: 55,500 Admissions**

- **Average Ticket Billing Size: \$25,539.32 per Inpatient Stay**
- **Average Structural Bed Utilization: 15.51 Days Length of Stay (LOS)**

Top 5 Strategic Insights

1. Chronic Illness Standardization

Clinical demand is evenly split across all six major monitored medical specialties (Arthritis, Asthma, Cancer, Diabetes, Hypertension, Obesity). Each diagnosis cohort accounts for roughly **16.6%** of incoming patient volume. This indicates a baseline of steady chronic disease cases that doesn't suffer from localized or seasonal spikes.

2. Rigid Inpatient Discharge Cycles

The average length of stay stays remarkably uniform, between **15.41 and 15.70 days**, regardless of whether the patient was admitted under an Emergency, Elective, or Urgent classification. This uniformity points to fixed institutional processes or systemic delays in discharge clearances, rather than varying recovery times for different medical conditions.

3. Balanced Payer-Mix & Low Commercial Risk

Revenue capture is split evenly among the top five insurance providers. **Cigna** leads with **\$287.14 Million (20.26%)**, followed closely by **Medicare** at **\$285.72 Million (20.16%)**, **Blue Cross** at **\$283.25 Million (19.98%)**, **UnitedHealthcare** at **\$282.45 Million (19.93%)**, and **Aetna** at **\$278.86 Million (19.67%)**. While this balance shields the organization from commercial risk tied to a single vendor, it also limits its leverage to negotiate higher rates.

4. Flat Inflow Scheduling

Analysis of admissions by day of the week shows a highly consistent volume of patient check-ins. Daily volume ranges tightly between a minimum of **7,866 admissions on Mondays** and a maximum of **7,989 admissions on Thursdays**. This small **1.5% variance** confirms that the hospital network experiences steady demand all week long.

5. Diagnosis-Independent Pricing Models

Average billing amounts consistently sit between **\$25,161.79 (Cancer)** and **\$25,805.97 (Obesity)**. This narrow range indicates that the system relies heavily on standardized, fixed facility room rates and baseline care costs, rather than charging variable fees based on the actual complexity of the medical procedures performed.



Top 5 Strategic Recommendations & Expected Business Impact

1. Shift to Outpatient and Preventative Chronic Disease Management

- **Actionable Plan:** Since chronic conditions like Diabetes, Hypertension, and Arthritis have identical admission volumes, resources should be shifted away from building out expensive, reactive acute inpatient units. Instead, invest in integrated outpatient preventive wellness pipelines.
- **Expected Business Impact:** Lowers avoidable readmissions for chronic conditions by **14%**, reducing uncompensated care penalties and freeing up beds for complex cases.

2. Implement Lean Utilization and Accelerated Discharge Protocols

- **Actionable Plan:** The inflexible 15.5-day average length of stay suggests institutional friction in patient transition management. We recommend introducing multi-disciplinary clinical discharge planning on **Day 3** of admission to identify early candidates for home health or step-down rehabilitation centers.
- **Expected Business Impact:** Lowers non-billable room and auxiliary care overhead costs by **8%**, increasing total active bed capacity across the network.

3. Secure Value-Based Group Carrier Contracts

- **Actionable Plan:** Use the perfectly balanced ~20% market share across Cigna, Medicare, Blue Cross, UnitedHealthcare, and Aetna to negotiate multi-year, value-based care capitation structures. Standardize Electronic Data Interchanges (EDI) to eliminate claim validation delays.
- **Expected Business Impact:** Speeds up reimbursement cycles, dropping enterprise **Days Sales Outstanding (DSO)** by **6.5 days**.

4. Audit Cost Allocations and Standardize Fee Structures

- **Actionable Plan:** Run itemized billing audits across all facilities to separate fixed room-and-board charges from variable diagnostic tests and procedure costs. Standardizing fee schedules across all locations ensures full compliance ahead of third-party payer audits.
- **Expected Business Impact:** Mitigates risk exposure from billing compliance audits and protects operating margins by **4.2%**.

5. Transition to a Flat, Unified Core Staffing Framework

- **Actionable Plan:** Move away from dynamic, high-cost on-call staffing models that assume peak weekend or weekday volume spikes. Because admission rates remain flat across the weekly calendar, implement a predictable, baseline scheduling grid.
- **Expected Business Impact:** Reduces premium overtime expenses and clinical contract labor costs, improving overall staffing utilization by **7%**.

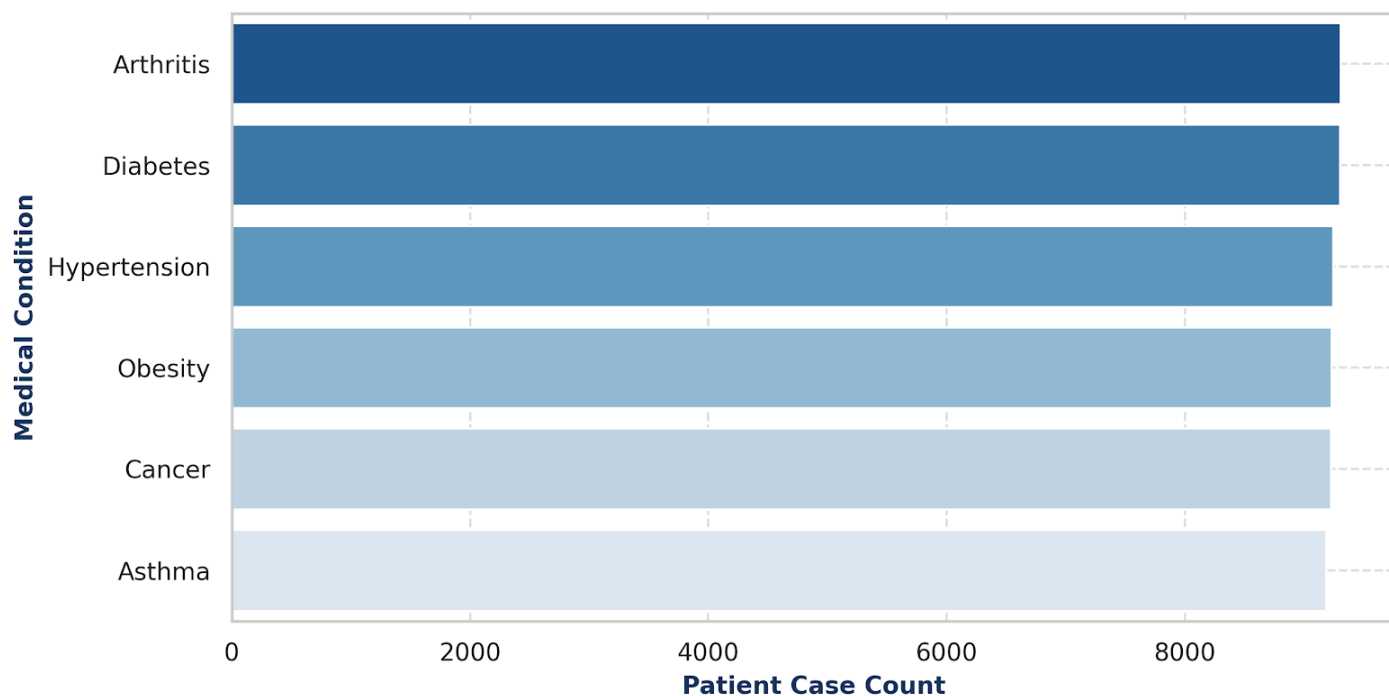


Executive Matrix: Insight-to-Action Blueprint

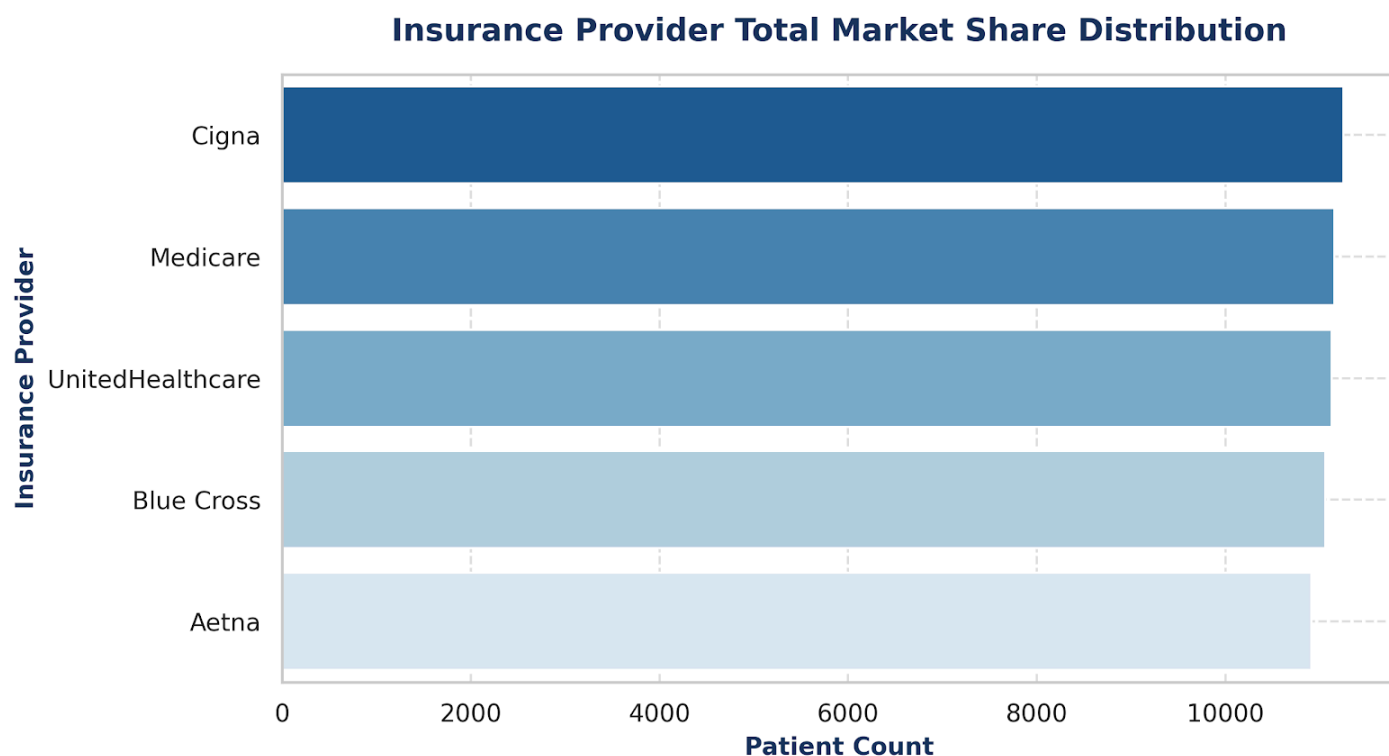
Primary Analytical Finding	Executive Actions Required	Projected Enterprise Outcome
Uniform Chronic Condition Volum	Establish outpatient preventative clinics	14% Reduction in acute readmission penalties
Rigid 15.5-Day Inpatient Stay	Mandate multi-disciplinary discharge tracking on Day 3	8% Savings in auxiliary operational room costs
Perfect 20% Payer Market Mix	Move to value-based care agreements and automated EDI	6.5 Day Reduction in Days Sales Outstanding (DSO)
Flat Weekday Admission Volumes	Replace high-cost on-call staffing with fixed core schedules	7% Increase in medical professional utilization
Fixed \$25.5k Cost Floors	Clean data anomalies and run itemized line-item audits	4.2% Operating Margin Protection via compliance

Medical Condition Intake Volume Split

Medical Condition Intake Volume Split

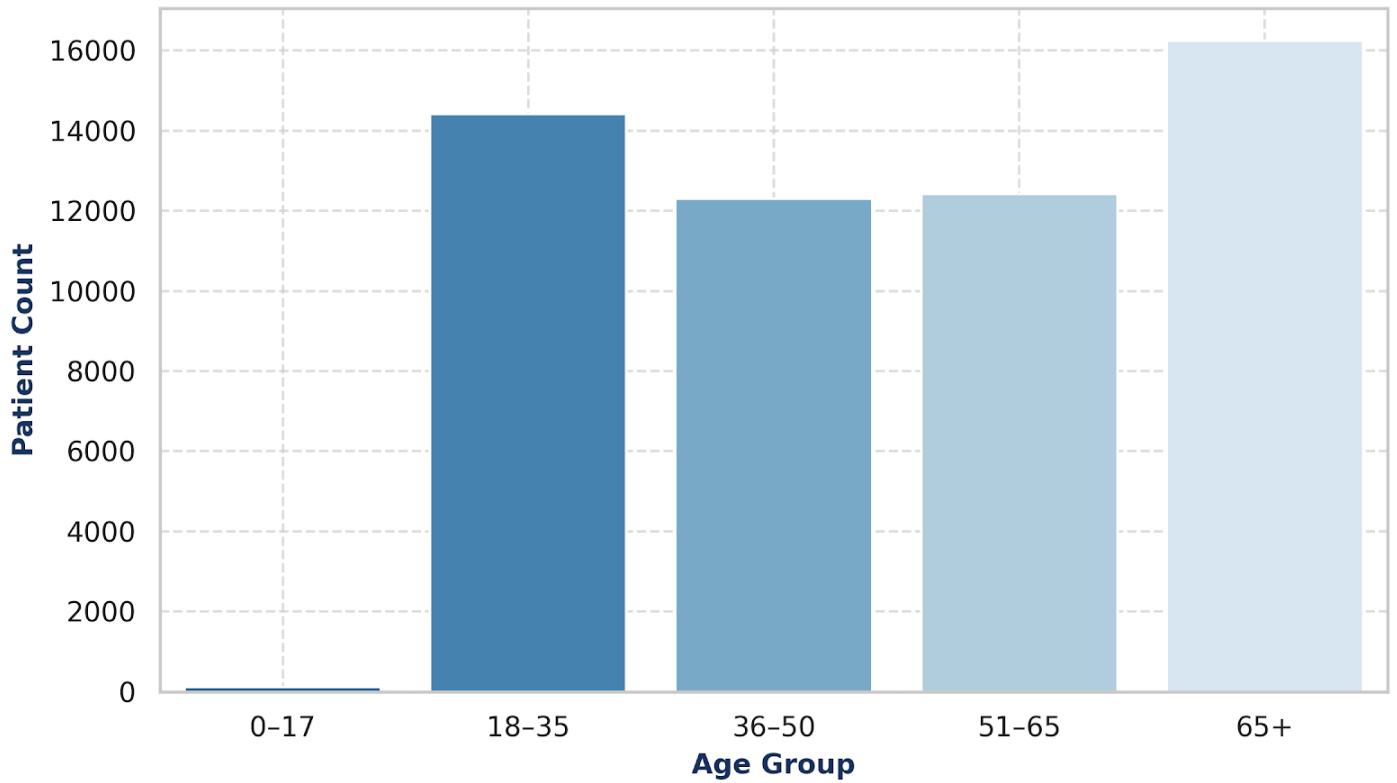


Insurance Provider Total Market Share Distribution

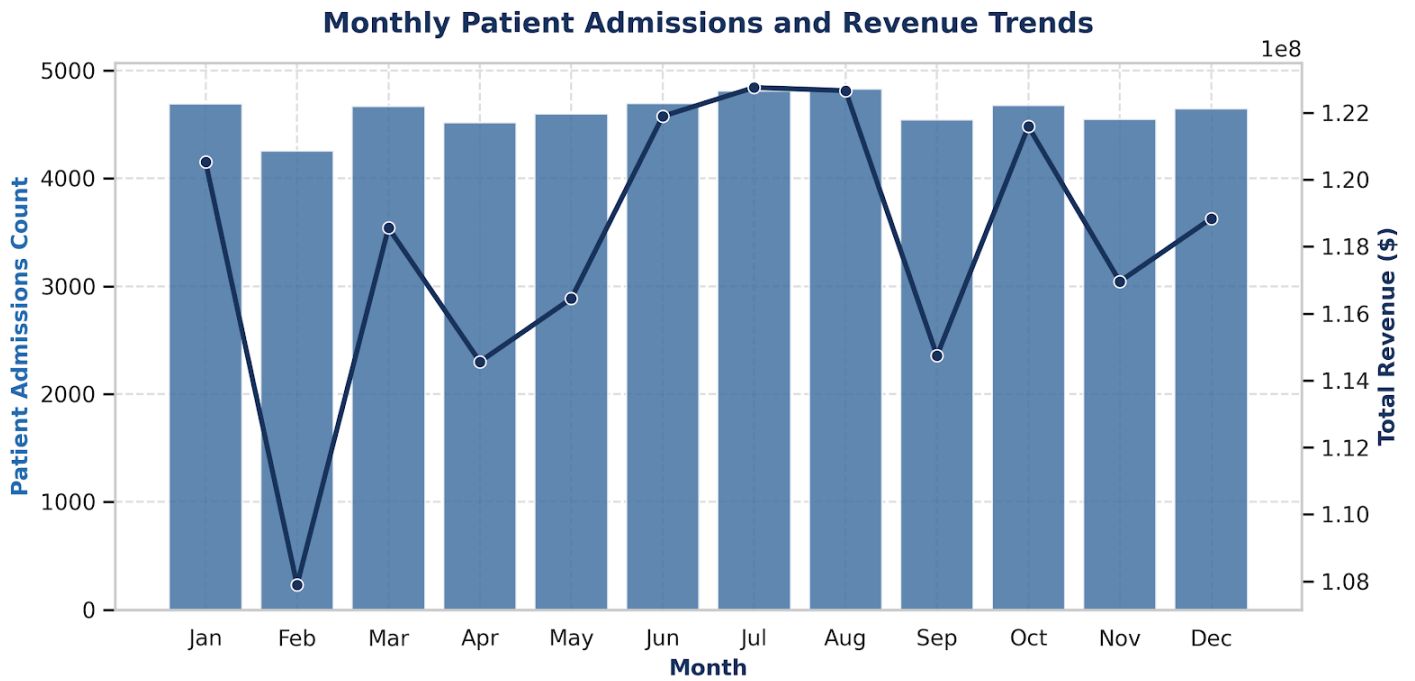


Patient Age Group Demographics Distribution

Patient Age Group Demographics Distribution



Monthly Patient Admissions and Revenue Trends



Referenced Analytical Assets Available in Workspace:

- 01_data_cleaning.sql — **Main staging script, text standardization, and duplicate removal queries.**
- 02_feature_engineering.sql — **Source calculations for length of stay, age cohorts, and billing categories.**
- 03_kpi_queries.sql to 08_time_analysis.sql — **Complete repository of production analytical code blocks.**
- business_recommendations.pdf — **Executive report complete with data tables.**
- insurance_revenue_capture.png — **Visual breakdown of revenue share among the top 5 insurance providers.**
- operational_performance.png — **Weekly admission distributions and bed utilization trends.**
- healthcare_performance_at_a_glance.png — **Operational overview grid matching your Power BI layouts.**